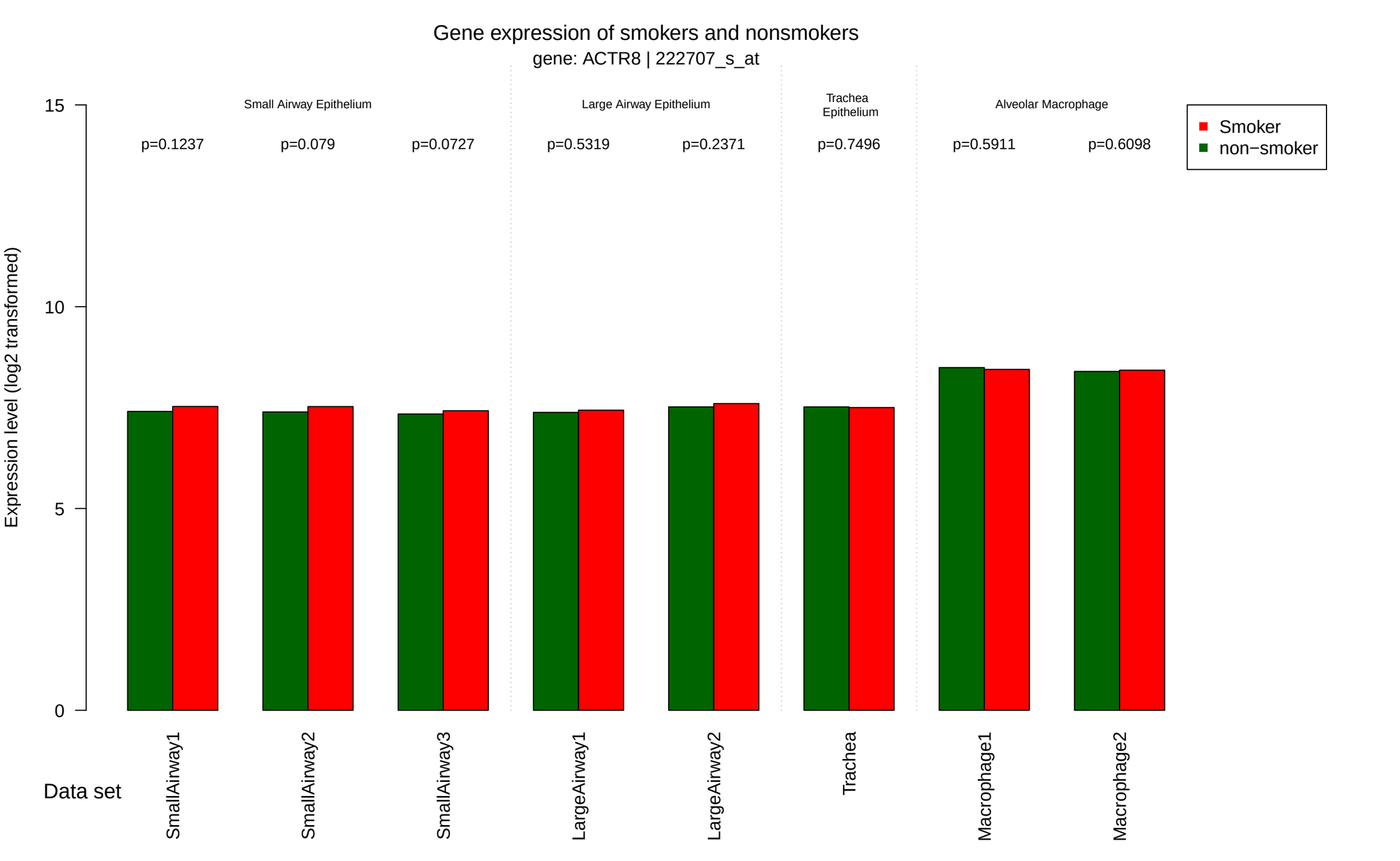
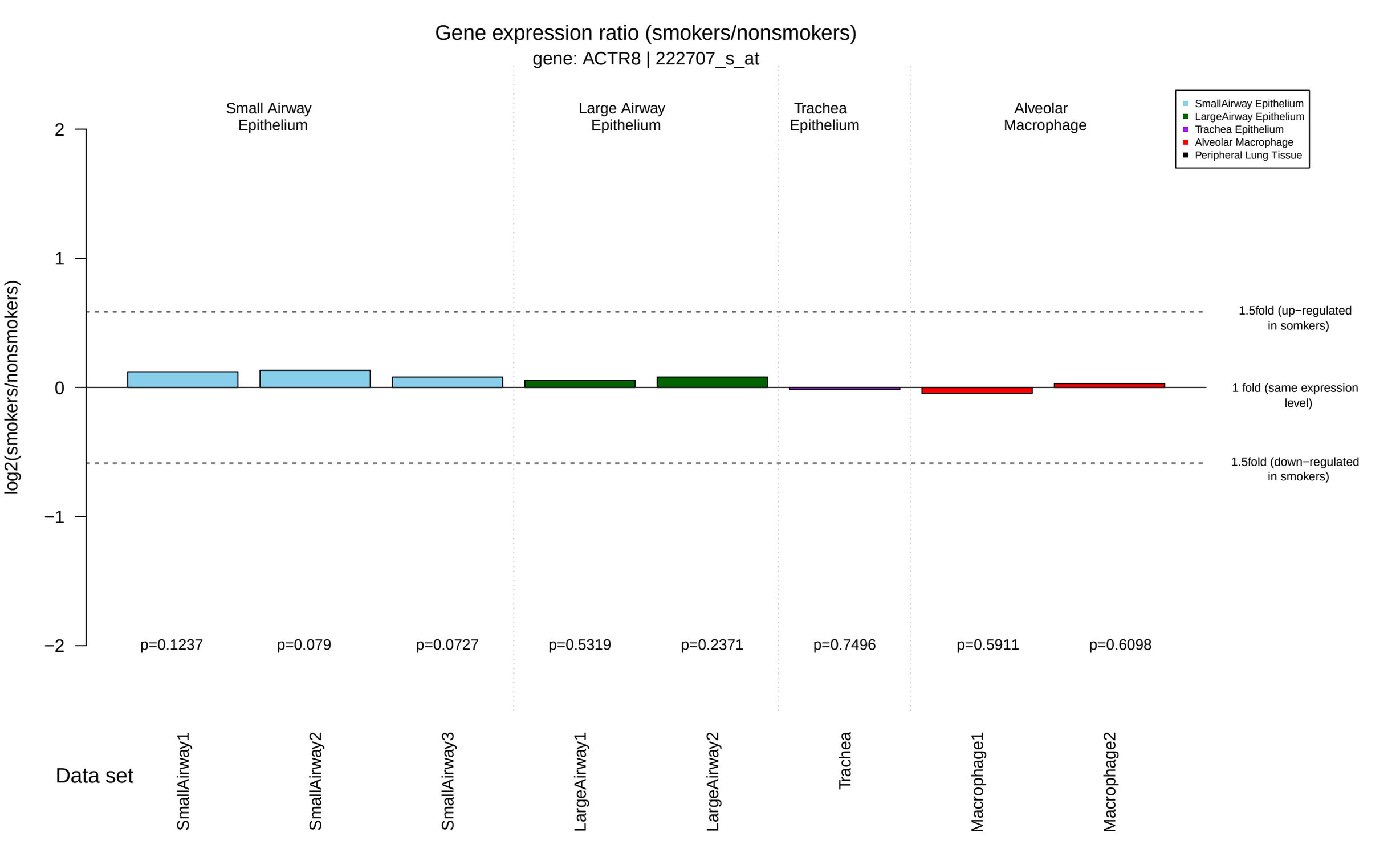
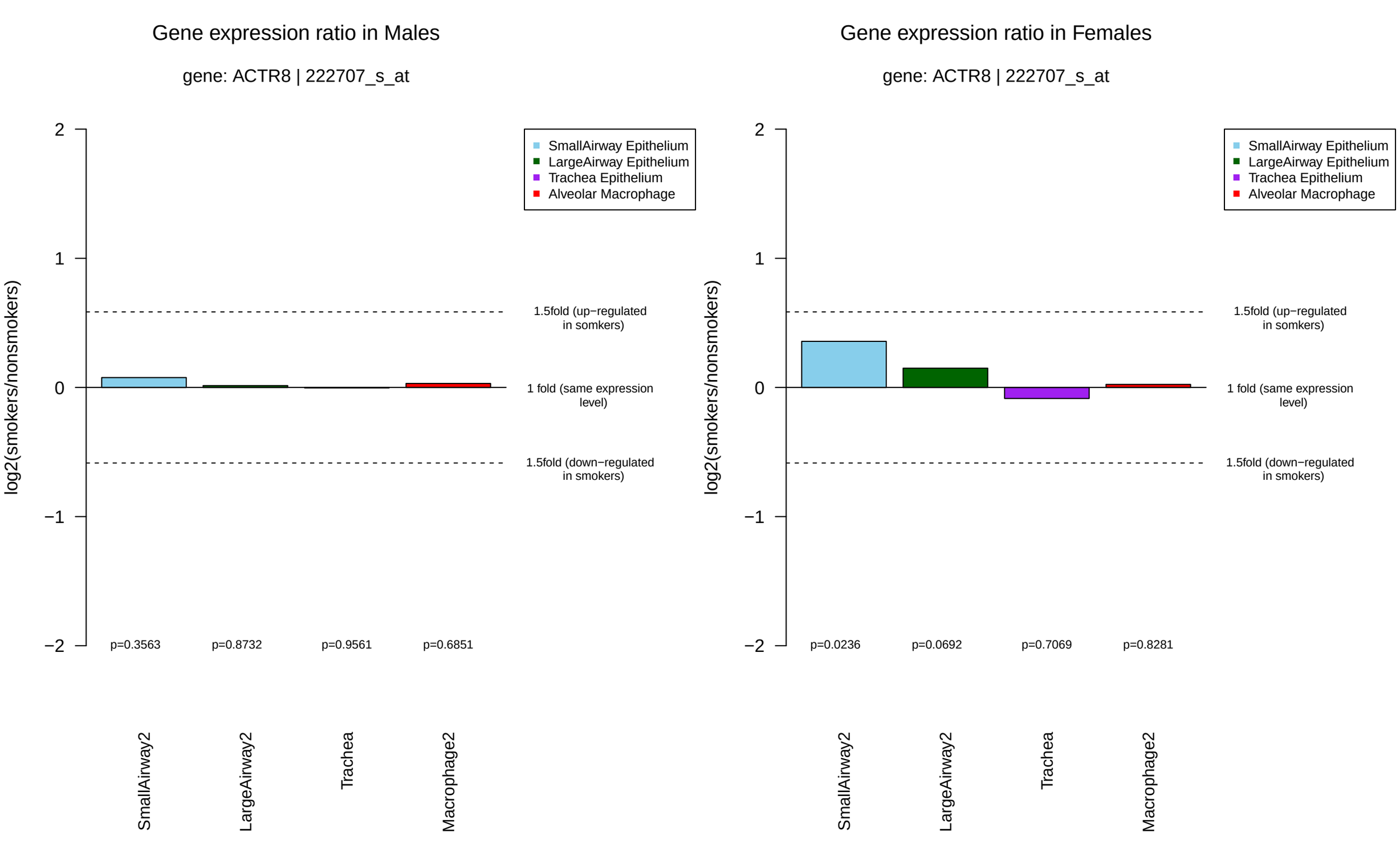
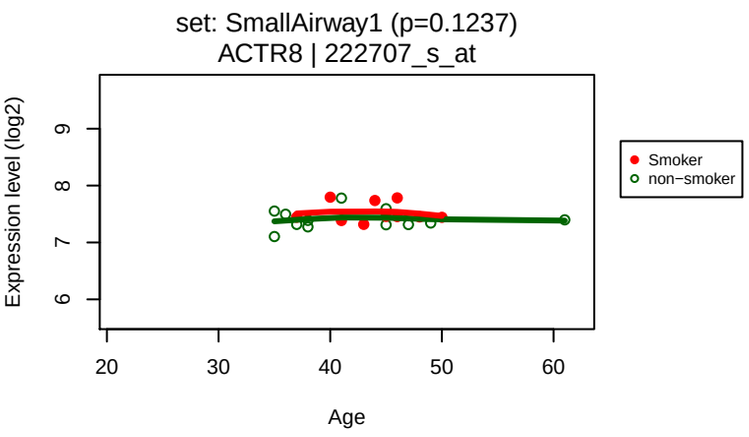
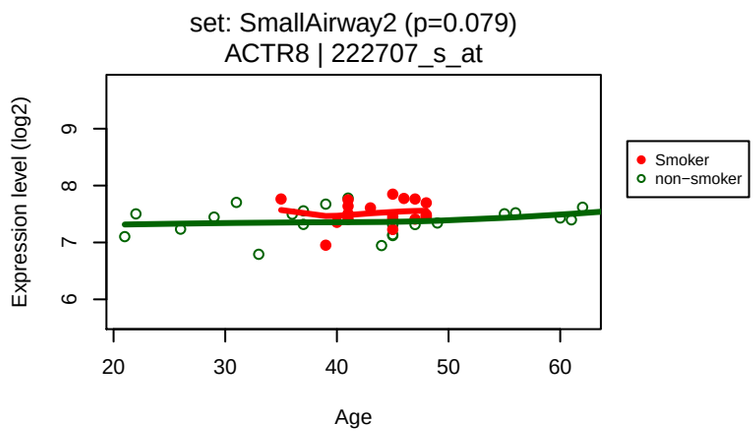
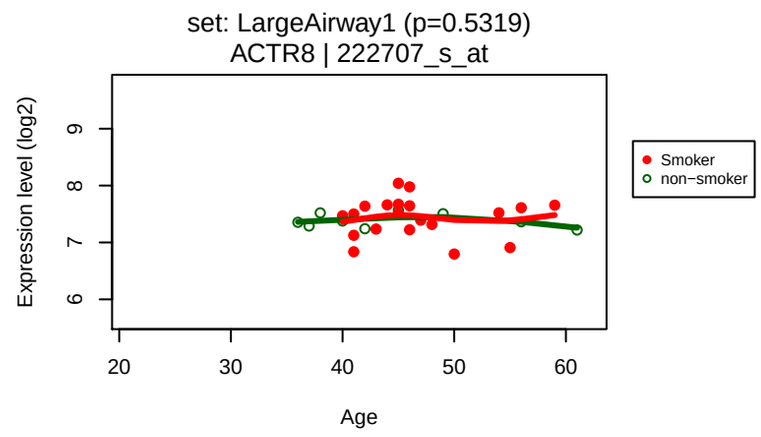

Gene expression of smokers and nonsmokers
gene: ACTR8 | 222707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1237
p=0.079
p=0.0727
p=0.5319
p=0.2371
p=0.7496
p=0.5911
p=0.6098
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

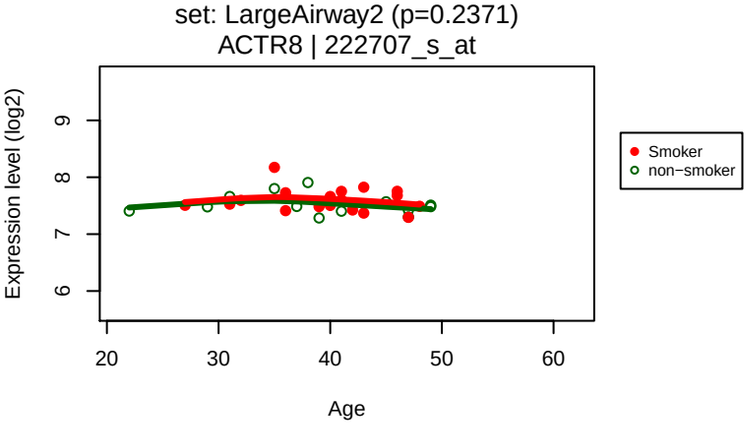

Gene expression ratio (smokers/nonsmokers)
gene: ACTR8 | 222707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1237
p=0.079
p=0.0727
p=0.5319
p=0.2371
p=0.7496
p=0.5911
p=0.6098
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

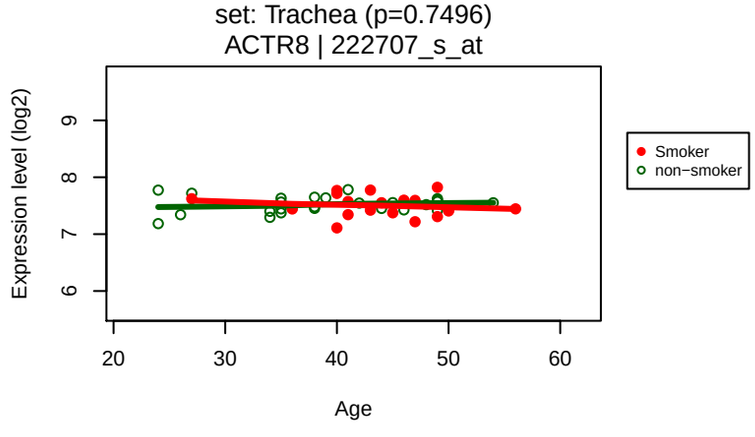

Gene expression ratio in Males
gene: ACTR8 | 222707_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3563
p=0.8732
p=0.9561
p=0.6851
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR8 | 222707_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0236
p=0.0692
p=0.7069
p=0.8281
SmallAirway2
LargeAirway2
Trachea
Macrophage2

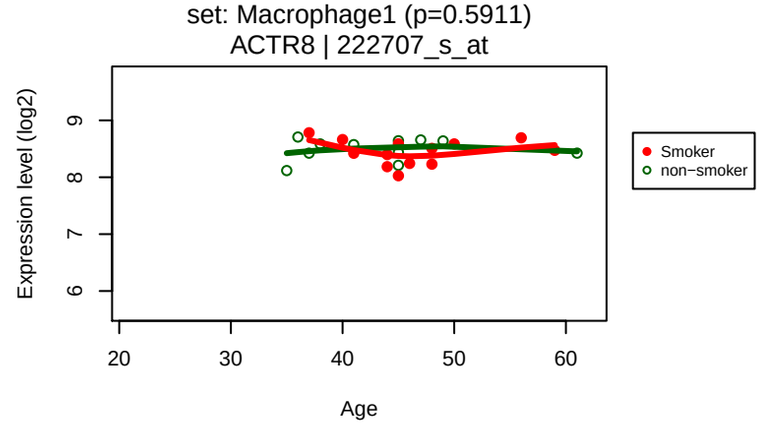

set: SmallAirway1 (p=0.1237)
ACTR8 | 222707_s_at
Expression level (log2)
Age
Smoker
non−smoker

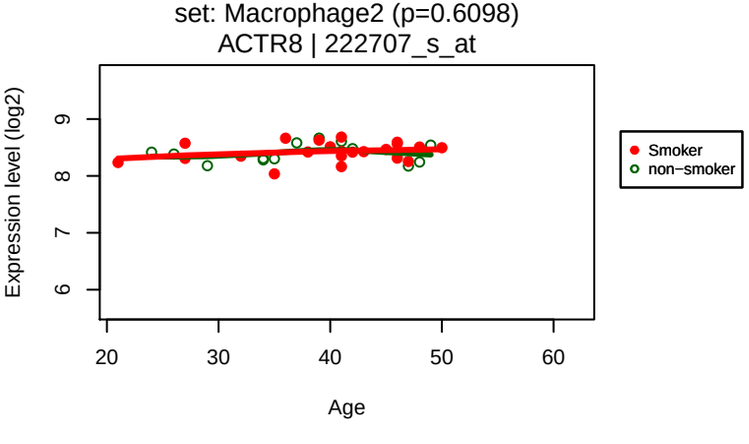

set: SmallAirway2 (p=0.079)
ACTR8 | 222707_s_at
Expression level (log2)
Age
Smoker
non−smoker

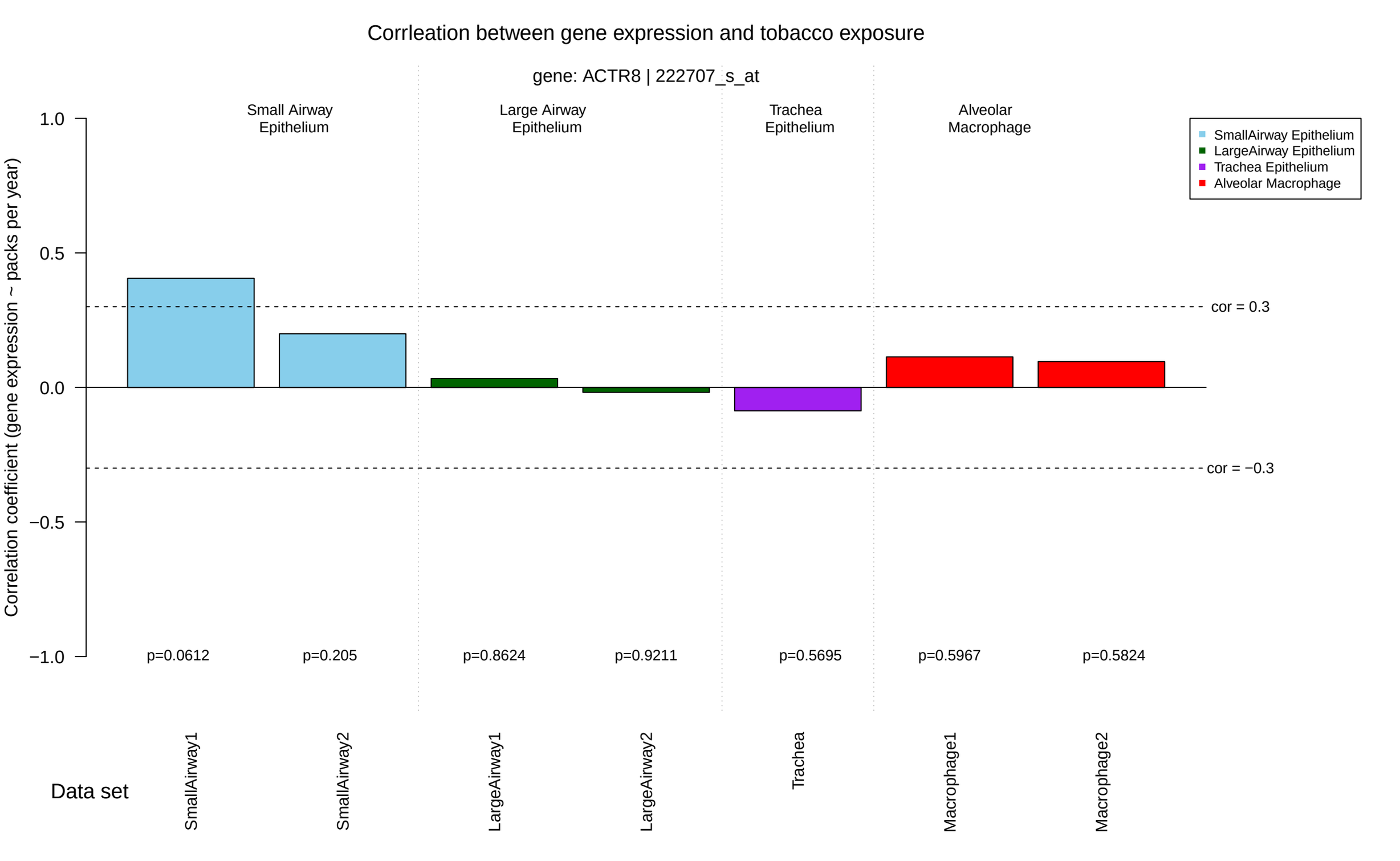

set: LargeAirway1 (p=0.5319)
ACTR8 | 222707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2371)
ACTR8 | 222707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7496)
ACTR8 | 222707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5911)
ACTR8 | 222707_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6098)
ACTR8 | 222707_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR8 | 222707_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0612
p=0.205
p=0.8624
p=0.9211
p=0.5695
p=0.5967
p=0.5824
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2